

Phase 5: Project Development

AI Comic Story Creator — SmartBridge Project Submission

1. Development Approach

Build the pipeline incrementally, agent by agent, so each stage can be validated independently before wiring the full orchestration: (1) script generation, (2) character anchor generation, (3) panel prompt generation, (4) image rendering, (5) compositing/export.

2. Core Modules

- **generate_comic_script()** — calls Gemini 1.5 Pro with a director-style system instruction and `response_mime_type='application/json'` to produce the structured script.
- **render_panel_image()** — combines art style, character anchor, action, and environment into a single prompt and calls Imagen 3 (`generate_images`) to synthesize each panel.
- **draw_speech_bubble()** — uses Pillow's `ImageDraw` to composite dialogue bubbles and captions onto a rendered panel at the coordinates the Panel Director specified.
- **ComicLayoutEngine** — calculates panel bounding boxes (x, y, w, h) for a given panel count at 300 DPI, handling margin and gutter spacing.

3. Reference Implementation Notes

The reference pipeline uses the google-genai SDK: a Gemini client generates the JSON script, Imagen 3 renders each panel from the combined visual-anchor prompt, and Pillow composites bubbles before the pages are assembled into the export format. See the project document / repo for the full code listing (script generation, image rendering, bubble compositing, and the orchestrator entry point).

4. Development Best Practices

- Validate every Gemini JSON response against the expected schema before passing it to the next agent.
- Log every Imagen prompt sent, for reproducibility and debugging character drift.
- Keep negative prompts standardized (e.g. 'text, watermark, photo-realistic, distorted faces, duplicate limbs, out of frame speech bubbles') to reduce unwanted artifacts.
- Implement retry/backoff for Imagen rate limits at the task-queue level, not inline in request handlers.